MIL-C-48596A(AR)
AMENDMENT 2
21 October 1993
SUPERSEDING
AMENDMENT 1
22 SEPTEMBER 1987

MILITARY SPECIFICATION

COMPUTER SYSTEM, BALLISTIC, M21: 11732979

This amendment forms a part of MIL-C-48596(AR), dated 30 June 1981, and is approved or use by the U.S. Army Armament, Munitions, and Chemical Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

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2.1, Inspection Equipment, Add "11744800 Reticle Projector (GFP)".

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3.3.23.2, Stabilization electronics interface signals, Subparagraph b, D (OUT) , Delete last sentence and substitute “The D (OUT) signal between W2P6-D(+) and W2P6-G(-) shall vary between zero and + 14-5 V dc when the RPU reticle deflects left and between zero and - 14.5 V dc when the RPU deflects right."

3.3.24.2, Built-in test, Delete and substitute “The built-in test circuitry shall be active with the LAMP-NORMAL-SYSTEM switch on the GCU in the SYSTEM position and shall isolate a fault in a defective unit and illuminate the appropriate fault light on the GCU for any value of cant within the limits of 10.0 , with an accuracy of 0.5. (See 4.6.2.24)"

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Table XIV, Test Equipment, Add: "11 11744800, Reticle Projector (GFP)”.

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* TABLE XV BCS hardware mechanism accuracy., Delete and substitute:

TABLE XV. BCS hardware mechanism accuracy.

Item	Ammo 1/	Range Man (km) 1/	Range Auto (V) 2/	BRL	Air Temp (°F) 1/	Alt (km) 1/	Rem. Tube Life (%) 1/	Crosswind (mph) Man 1/	Crosswind (mph) Auto 3/	Cant Angle 0.05	LL 4/	Turret Rate Tach mil/s 0.1 %	Turret Rate Stabil (V) 0.007 4/	MT 1/	Stab 5/	Elevation OU Dial (mil)	Deflection RPU (mil) 6/	Deflection Laser DL FDBK (V ac) 7/
1	APDS M392	--	1.968	OFF	+100	2.5	80	-	0	- 10	1	--	-1.104	ON	ON	4.30	-19.76	1.175
		8/														0.20	0.35	0.022
2	APDS M728	--	3.723	OFF	+100	2.5	80	+10	-	0	0	0	--	OFF	OFF	9.55	- 0.16	0.028
																0.25	0.25	0.016
3	HEAT M456	--	1.277	OFF	+100	2.5	80	-	0	- 10	1	- 20	--	OFF	OFF	9.00	22.12	-1.304
																0.25	0.53	0.034
4	HEAT M456	--	1.277	ON	+100	2.5	80	-	0	- 10	1	- 20	--	OFF	OFF	9.00	22.12	-1.282
																0.25	0.53	0.034
5	HEP M393	2.85	-	OFF	+100	2.5	80	-	0	0	0	0	--	OFF	OFF	45.25	- 1.73	0.119
																1.25	0.33	0.021
6	HEP M393	--	1.277	OFF	+100	2.5	80	-	0	+ 10	1	+ 5	--	OFF	OFF	15.35	- 7.58	0.446
																0.35	0.42	0.027
7	FSDS M735	--	3.723	OFF	+100	2.5	80	+10	-	+ 10	0	0	--	OFF	OFF	7.95	1.47	-0.368
																0.25	0.25	0.016
8	FSDS M735	--	0.934	OFF	+100	2.5	80	-	0	0	1	--	+1.104	ON	ON	2.10	8.90	-0.538
																0.20	0.30	0.019
9	HEAT M456	--	3.032	OFF	+100	2.5	80	+40	-	0	0	0	--	OFF	OFF	14.50	6.92	-0.391
																0.30	0.38	0.024
10	HEP M393	--	1.968	OFF	+100	2.5	80	+40	-	0	0	0	--	OFF	OFF	23.30	7.35	-0.423
																0.45	0.42	0.027
11	HEP M393	--	1.968	OFF	+100	2.5	80	-	0	0	0	0	--	OFF	OFF	23.30	- 0.94	0.065
																0.45	0.33	0.019
12	HEAT M456	--	3.032	OFF	STD	0	NEW	-	-40	5.0	0	0	--	OFF	OFF	18.55	8.39	-0.478
										0						0.30	0.70	0.043
13	HEAT M456	--	3.032	OFF	STD	0	NEW	-	-40	5.0	0	0	--	OFF	OFF	18.55	8.79	0.534
										0						0.30	0.70	0.043
14	HEP M393	--	1.968	OFF	STD	0	NEW	-	+20	2.5	0	0	--	OFF	OFF	26.55	3.32	-0.186
										0						0.45	0.62	0.038

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TABLE XV. BCS hardware mechanism accuracy. - Continued

Item	Ammo 1/	Range Man (km) 1/	Range Auto (V) 2/	BRL	Air Temp (°F) 1/	Alt (km) 1/	Rem. Tube Life (%) 1/	Crosswind (mph) Man 1/	Crosswind (mph) Auto 3/	Cant Angle 0.05	LL 4/	Turret Rate Tach mil/s 0.1 %	Turret Rate Stabil (V) 0.007 5/	MT 1/	Stab 5/	Elevation OU Dial (mil)		Deflection RPU (mil) 6/		Deflection Laser DL FDBK (V ac) 7/	
15	HEP M393	--	1.968	OFF	STD	0	NEW	-	-20 2.5	0	0	0	--	OFF	OFF	26.55	0.45	- 5.44	0.62	0.330	0.038
16	HEP M393	--	1.968	OFF	STD	0	NEW	-	0	0	0	0	--	OFF	OFF	26.55	0.45	- 1.06	0.33	0.072	0.021
17	APDS M392	--	3.032	OFF	+ 80	3.0	80	+5	-	- 5	1	- 10	--	OFF	OFF	8.70	0.25	19.78	0.47	-1.149	0.029
18	APDS M728	--	1.968	OFF	- 20	2.0	20	-5	-	+ 5	1	+ 10	--	OFF	OFF	6.05	0.25	-14.13	0.35	0.842	0.022
19	HEAT M456	--	0.904	OFF	+120	1.0	NEW	+10	-	- 5	1	+ 5	--	OFF	OFF	3.35	0.25	- 3.67	0.32	0.202	0.021
20	HEP M393	--	1.968	OFF	- 40	-0.1	50	-10	-	+ 5	0	0	--	OFF	OFF	30.30	0.60	- 0.98	0.33	0.067	0.021
21	FSDS M735	--	0.904	OFF	- 50	1.0	80	+10	-	- 5	1	- 20	--	OFF	OFF	3.15	0.25	11.72	0.35	-0.704	0.021

1/ Dialed-in inputs.

2/ Simulated range input. V ac 0.005 V. Table II, item 2.1.1 (1.064 V/km).

3/ Crosswind sensor connected in still air when zero wind is specified. Crosswind sensor oriented at full crosswind. Plus (+) value indicates wind from the right. Minus (-) value indicates wind from the left. See Figure 2.

4/ See Table II. Item 2.2.5 equals zero. Itme 2.2.6 equals one. The rate inputs must be present at a constant value for a period of not greater than 1.5 seconds before Lead Lock signal is set to one.

5/ See Table II, items 2.2.1 through 2.2.4.

6/ Positive voltage indicates the output is in phase with AC Ref 1. Negative voltage indicates the output is out of phase with AC Ref 1.

7/ RPU solutions are given as observed through the gunner's sight. For actual RPU displacement, multiply solution by 1.464.

8/ A dash (-, --) indicates the alternate input is used.

9/ Nominals and tolerances premised on GCU knobs in nearest detent position.

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